

VibeValue: Language Model for Enhanced Financial Sentiment Insights

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Abstract

Pretrained language models with contextual understanding have achieved notable advancements in diverse NLP tasks through training on extensive amounts of unlabeled text. The financial sector generates substantial volumes of financial communication text, yet there remains a lack of pretrained language models specifically tailored for this domain. In this study, we introduce VibeValue, a sentiment analysis model fine-tuned on finance domain-specific data using a transfer learning approach built on FinBERT. VibeValue achieves an accuracy of 98.01% and an F1-score of 0.973 on the Financial PhraseBank dataset with 100% annotator agreement, outperforming existing state-of-the-art models including FinBERT, ULMFiT, and LSTM-based approaches across all evaluation metrics. These results demonstrate the effectiveness of domain-specific pretraining and fine-tuning for financial sentiment classification.

Introduction

Financial markets are dynamic ecosystems where billions of transactions are initiated and completed daily, making them a focal point of global economic activity. Prices in an open market reflects all of the available information regarding assets exchanged in an economy.(Malkiel 2003) The ability to comprehend and respond to the intricate interplay of factors influencing these markets is a fundamental challenge for investors, traders, and financial institutions. Analysis of financial texts, be it news, analyst reports or official company announcements is a possible source of new information with an unprecedented amount of such text being created every day, manually analyzing these and deriving actionable insights from them is too big of a task for any single entity. Hence, automated sentiment or polarity analysis of texts produced by financial actors using natural language processing (NLP) methods has gained popularity during the last decade.(Guo, Shi, and Tu 2016) Sentiment analysis is the task of extracting sentiments or opinions of people from written language.(Liu 2012)The two paradigms of recent effort are:

1. Machine learning methods with features extracted from text with "word counting" (Agarwal and Mittal 2016; Araci 2019; Tripathy, Agrawal, and Rath 2016; Whitelaw, Garg, and Argamon 2005).
2. Deep learning methods, where text is represented by a sequence of embeddings (Araque et al. 2017; Severyn and Moschitti 2015; Zhang, Wang, and Liu 2018). The former suffers from inability to represent the semantic information that results from a particular sequence of words, while the latter is often deemed as too "data-hungry" as it learns a much higher number of parameters (Araci 2019).

¹ Identify applicable funding agency here. If none, delete this.

VibeValue builds on BERT (Devlin et al. 2018) but trains on financial communication data specifically. That pretraining is what separates it from a generic language model the jargon, market-specific phrasing, and contextual nuances that general models routinely miss are exactly what VibeValue was exposed to during training. The result is a model that doesn't need to approximate financial language because it was built on it.

BERT is what VibeValue is built on, and the choice matters. Earlier approaches fixed word embeddings, bag-of-words treated words as isolated units. "Bearish" means the same thing regardless of what surrounds it, which works until it doesn't. BERT reads the entire sentence before assigning meaning to any part of it, so context shapes the representation rather than being ignored. For financial text, that difference is significant. Sentiment in market language rarely lives in a single word it sits in how terms relate to each other, and a model that can't track those relationships will miss it.

The choice to utilize VibeValue for market sentiment prediction over traditional methods is motivated by several compelling reasons. Traditional approaches, such as carefully crafted financial sentiment lexicons like Loughran and McDonald (2011)(Loughran and McDonald 2011), may seem a viable solution because they incorporate existing financial knowledge into textual analysis. Conversely, these approaches rely on "word counting" techniques, which are inadequate for a comprehensive analysis of a text's semantic depth. Such methods are limited in their ability to discern the nuances inherent in financial discourse and the contextual variability of sentiment. VibeValue handles financial text better than traditional sentiment tools for a concrete reason it was pretrained on financial data, not adapted to it after the fact. That distinction shows up in how it processes industry terminology, market slang, and the kind of figurative language that turns up in analyst reports and earnings calls. A rule-based system has to be told what those mean. VibeValue picks them up from context. Traditional methods also age poorly. Market language shifts, new terminology emerges, and rule-based systems need manual updates to keep pace. A deep learning model adjusts through continued training rather than hand-coded fixes, which makes it more practical to maintain as language and market conditions evolve.

Financial text is harder to read than it looks. The language is dense, sentiment shifts fast, and the margin for misclassification carries real consequences for investors and analysts working off that data. VibeValue was built to address that gap. Existing models struggle with the nuances of financial language accuracy, precision, recall, and F1 scores tend to fall short when models aren't specifically adapted for the domain. Finetuning on financial data isn't optional here, it's the whole point. The sections that follow cover the methodology, results, and what the numbers actually mean in practice. The core argument is straightforward: a model calibrated for financial text will outperform a general-purpose one on this task, and the difference shows up where it matters most in the metrics that reflect real classification quality, not just surface-level accuracy.

Related Work

This section reviews the key architectures and approaches that form the foundation of modern NLP-based sentiment analysis. We examine sequential models such as LSTM and

ELMo, transfer learning frameworks like ULMFiT, and attention-based architectures including Transformers and BERT, which collectively motivate the design choices behind VibeValue.

LSTM

Long Short-Term Memory (LSTM) networks are a specialized variant of recurrent neural networks engineered to persist long-term dependencies across sequences. They achieve this via gate mechanisms, such as the **discard gate**, which determines which data within the cell state is retained or purged. It processes the preceding hidden state h_{t-1} and the current input x_t , yielding a value between 0 and 1 for each element in the state S_{t-1} . Mathematically:

$$g_{f,t} = \sigma(\Omega_f \cdot [h_{t-1}, x_t] + \theta_f)$$

The **selection gate** identifies which new data should be integrated into the cell state. This involves a sigmoid layer to choose update values and a tanh layer to generate a vector of prospective candidate values.

Mathematically:

$$g_{i,t} = \sigma(\Omega_i \cdot [h_{t-1}, x_t] + \theta_i)$$

$$\tilde{S}_t = \tanh(\Omega_s \cdot [h_{t-1}, x_t] + \theta_s)$$

The **state transition** merges the outputs of the discard and selection gates to transform the previous state S_{t-1} into the current state S_t .

Mathematically:

$$S_t = g_{f,t} \odot S_{t-1} + g_{i,t} \odot \tilde{S}_t$$

LSTMs serve as core architectures for sequential modeling, including financial market trends and natural language processing. In linguistic tasks, a critical step is the initial encoding of tokens. A standard strategy involves utilizing pre-trained weights, such as **GLoVe** (Merity, Keskar, and Socher 2017), which generates word embeddings via unsupervised learning on a log-bilinear regression model using word-word co-occurrence data. While GLoVe captures semantic associations where similarity between vectors v_i and v_j is determined by:

$$\text{sim}(v_i, v_j) = \frac{v_i \cdot v_j}{\|v_i\| \|v_j\|}$$

It remains limited as it cannot adjust representations based on the specific sequence context, highlighting the necessity for advanced context-aware models.

ELMo

Static embeddings give every word one fixed vector "bank" means the same thing whether you're talking about a river or a loan. ELMo (Peters et al. 2018) dropped that assumption. It

stacks bidirectional LSTMs to read sequences from both ends, so each token picks up a representation from its actual surroundings. The model trains on token sequence probabilities across a target vocabulary, which is what grounds those context-sensitive representations in real language patterns.

The specific word-level embeddings are derived as follows:

$$\text{ELMo}(\tau_t) = \Gamma \cdot (\lambda_{token} \cdot [\vec{\psi}_t; \tilde{\psi}_t])$$

where Γ represents a linear transformation matrix and λ_{token} is a vector of trainable weights.

Specifically, ELMo assesses the probability of a token by considering both the antecedent and subsequent context in a sequence. The architecture dynamically tunes the weights for representations across various LSTM levels to produce a unified, context-aware vector for every token. These enriched representations function as high-quality initializations for a variety of subsequent NLP objectives.

ELMo extracts contextual data by synthesizing the full sequence of hidden states $\vec{\psi}_1, \vec{\psi}_2, \dots, \vec{\psi}_T$ and $\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_T$ through a layer-wise aggregation process. Mathematically, the comprehensive contextual embedding for an input sequence X is defined as:

$$\text{ELMo}(X) = \Gamma \cdot \left(\sum_{k=1}^L \lambda_{context}^k \cdot [\vec{\psi}_{t,k}; \tilde{\psi}_{t,k}] \right)$$

where $\lambda_{context}^k$ denotes the learned parameters for each specific layer k .

ULMFiT

ULMFiT (Howard and Ruder, n.d.) is a transfer learning framework for NLP that takes a different approach from ELMo rather than freezing the pretrained model and only training task-specific layers on top, it finetunes the entire language model jointly with those layers.

The backbone is AWD-LSTM (Devlin et al. 2018), an LSTM variant that uses aggressive dropout regularization to keep the model from overfitting during finetuning. For sentiment classification, two fully connected layers sit on top of the pretrained AWD-LSTM. The first takes the pooled hidden representations $\mathbf{h}_{pooled}$ as input and does the bulk of the classification work.

Given a sequence of input tokens $s_1, s_2, \dots, s_T$ and their corresponding targets $z_1, z_2, \dots, z_T$, the training objective is to minimize the negative log-likelihood (cross-entropy) risk:

$$\mathcal{L}_{LM} = - \sum_{t=1}^T \ln P(z_t | s_1, s_2, \dots, s_{t-1})$$

ULMFiT (Howard and Ruder, n.d.) handles domain adaptation and task finetuning through a few deliberate training choices. The most distinctive is **slanted triangular learning rates** (STLR) the step size α climbs linearly through an early portion of training, then drops

off exponentially. The shape of that schedule is what drives faster convergence without sacrificing generalization.

The learning rate trajectory $\alpha(t)$ is formally defined as:

$$\alpha(t) = \frac{\alpha_{peak} - \alpha_{base}}{p \cdot N} \cdot \min(t, p \cdot N - t) + \alpha_{base}$$

where α_{peak} is the maximum rate, α_{base} is the minimum rate, N is the total iterations, and p represents the fraction of steps used for the linear increase.

Transformers

The Transformer is an attention-based architecture built for sequential data, and a practical alternative to recurrent networks (Kraus and Feuerriegel 2017). It was originally designed as a sequence-to-sequence model with both encoder and decoder components the focus here is on the encoder, which is where most of the relevant processing happens for our purposes. Each encoder layer runs the same process: compute attention scores from query, key, and value vectors via dot product, scale them, apply softmax, then produce a weighted sum of the value vectors. Tokens that score higher pull more weight in the final representation the model is essentially deciding, at each layer, which parts of the input are worth paying attention to.

Mathematically:

- Attention Score:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)$$

- Output:

$$Attention(Q, K, V) \cdot V$$

paired with a fully connected feed-forward network which applies two linear transformations separated by a non-linear activation function, usually ReLU.

Mathematically:

$$FFN(x) = ReLU(xW_1 + b_1)W_2 + b_2$$

Within each self-attention layer, the model learns three mappings from the input embeddings: key, query, and value. For each token, a similarity score is calculated by taking the dot product of its query vector and the key vectors of all other tokens. These scores then determine how much each value vector contributes, which creates a refined representation for each token.

Multi-headed attention runs this process several times in parallel, each head attending to different aspects of the sequence. The outputs are then passed through shared fully connected layers, giving the model another pass at capturing relationships the attention

heads surfaced. As Vaswani et al. (Vaswani et al. 2017) argued, this is where Transformers have a clear edge over RNN-based models. RNNs process sequences step by step, which makes GPU parallelization difficult and causes information from distant tokens to degrade over many steps. Transformers sidestep both problems every token attends to every other token directly, regardless of distance.

BERT

BERT (Devlin et al. 2018) dropped the unidirectional assumption that earlier models like ELMo and AWD-LSTM relied on. Rather than predicting words in sequence, it masks roughly 15% of input tokens and trains the model to reconstruct them pulling context from both directions simultaneously, not just what came before. This forces it to draw on context from both directions at once, not just what came before. Alongside masking, BERT trains on **Next Sentence Prediction** (NSP), learning to identify whether two sentences actually follow each other. That second objective is what gives the model a handle on inter-sentence relationships, not just word-level patterns. The input architecture merges token, segment, and positional embeddings, utilizing unique indicators like *[CLS]* for classification pooling and *[SEP]* for boundary demarcation. BERT is distributed in two primary scales: *BERT-base* (12 layers, 768 hidden units, 110M parameters) and *BERT-large* (24 layers, 1024 hidden units, 340M parameters), both pretrained on massive corpora including Wikipedia and BookCorpus (Zhu et al. 2015).

Proposed Model

This study introduces **VibeValue**, a sentiment model built specifically for financial text. General-purpose models struggle with financial language the vocabulary is specialized, sentiment shifts quickly, and the same phrase can carry opposite implications depending on market context. VibeValue was built to handle that.

It builds on **FinBERT** (Araci 2019; Yang et al. 2020), a BERT-based model already oriented toward financial language, which gave VibeValue a more relevant starting point than a general-purpose baseline. Training followed a transfer learning approach (Howard and Ruder, n.d.; Peters et al. 2018) the model first learned from large unlabeled datasets, then those weights were carried over and refined for the specific task. In finance, where cleanly annotated data is hard to come by, that approach is practical in a way that training from scratch simply isn't.

VibeValue was pretrained on a financial corpus before any finetuning happened. That extra step matters financial language has its own patterns, terminology, and implicit meanings that a general-purpose model picks up slowly if at all. Pretraining on domain-specific text gives the model a better starting point.

Some structural changes were made to the attention heads and output layers to improve how the model handles sentiment classification specifically. From there, finetuning on annotated financial datasets helped it adapt across different text types the language in a

news headline reads differently than an analyst report, and the model needed exposure to both. The result is a model that's calibrated for financial text rather than just tolerant of it.

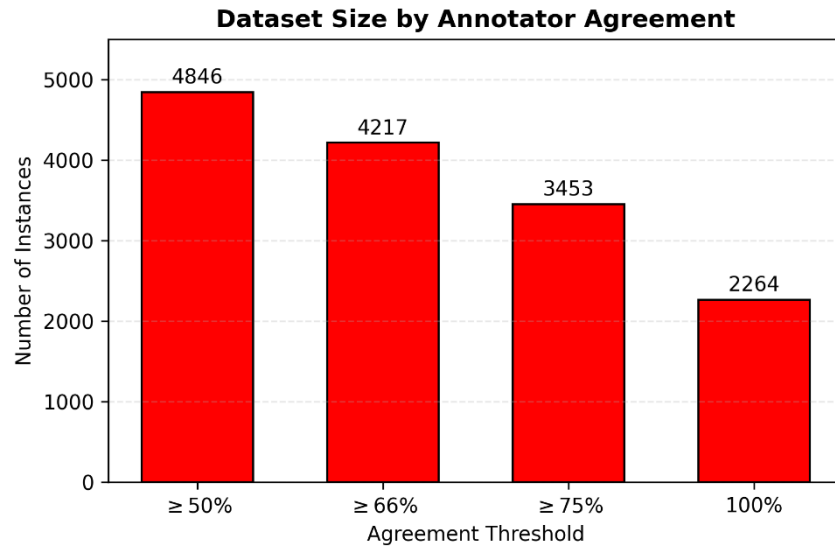
Module Design and Implementation

The implementation of our project VibeValue a sophisticated sentiment analysis model tailored for the financial domain employs a pre-trained BERT model on financial datasets comprising of a huge financial corpus to introduce domain specific vocabulary of the finance sector and using it as our base model on which we employ our fine-tuning strategies by setting up and optimizing the hyper parameters and selecting specific combinations of datasets to employ transfer learning. The process of fine-tuning our model on different annotated datasets for fine-tuning the classification layer for sentiment analysis viz. Financial PhraseBank (Malo et al. 2013), a combined dataset of FiQA(Maia et al. 2018) and Financial PhraseBank. Our model is designed to understand the subtleties and context of financial language, which improves its capacity to accurately gauge market sentiment. Beyond fine-tuning, we've optimized hyperparameters and employed methods to address imbalanced datasets, guaranteeing strong performance. This model is the backbone of VibeValue, offering a potent tool for real-time sentiment analysis in financial markets. It enables users to make informed choices in a landscape where sentiment is often the deciding factor.

Datasets

Financial PhraseBank

The main sentiment analysis dataset used to train our model is Financial PhraseBank from Malo et al. 2014(Malo et al. 2013). This release of the financial phrase bank covers a collection of 4840 sentences. The selected collection of phrases was annotated by 16 people with adequate background knowledge on financial markets. Three of the annotators were researchers and the remaining 13 annotators were master's students at Aalto University School of Business with majors primarily in finance, accounting, and economics. Given the large number of overlapping annotations (5 to 8 annotations per sentence), there are several ways to define a majority vote-based gold standard. To provide an objective comparison, we have formed 4 alternative reference datasets based on the strength of majority agreement:



Impact of annotator agreement thresholds on the available sample size of the Financial PhraseBank dataset.

A combined Dataset of FiQA and Financial PhraseBank

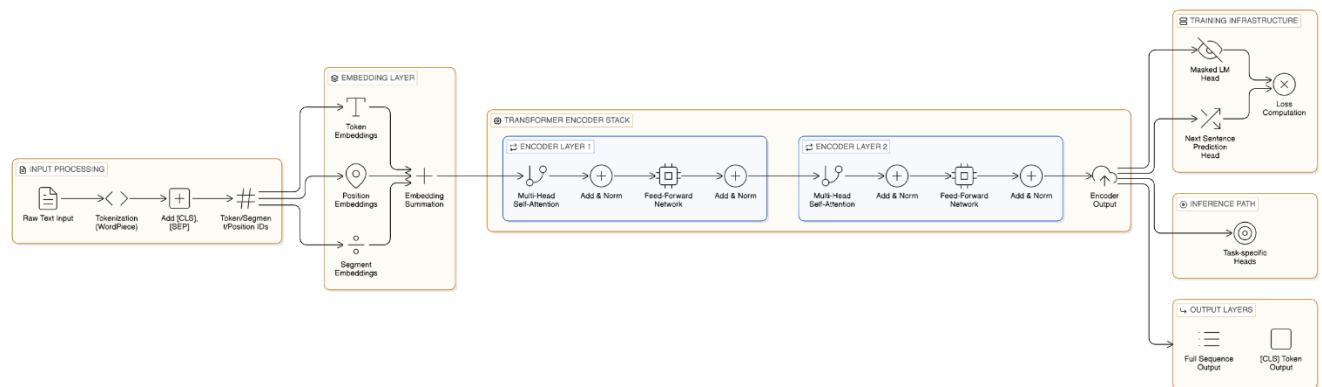
FiQA(Maia et al. 2018) is a dataset that was created for WWW '18 conference financial opinion mining and question answering challenges. The Dataset that we have used has 5843 annotated sentences which is used to train the model.

Model Architecture

Data Collection

Financial PhraseBank (Malo et al. 2013), assembled by Malo et al. (2014), was the starting point for finetuning. The dataset pulls 4845 English sentences from financial news sourced via LexisNexis real reporting, not synthetic examples. Sixteen experts with finance and business backgrounds went through each sentence and labeled it based on how they thought it would affect company stock prices. 20% of the data was set aside for testing, which left 3101 sentences to train on. VibeValue was finetuned on this split first, before anything else the foundation everything subsequent builds on.

Layers and components relevant to sentiment analysis



VibeValue's architecture is predicated on BERT, which employs a sequential stack of twelve Transformer encoder layers. Each of these layers processes its input via two primary components: a multi-headed self-attention mechanism and a feed-forward network. This layered design facilitates a more nuanced comprehension of textual data, as opposed to a single-pass processing of the entire input.

The input layer handles tokenization using BERT's WordPiece approach, which breaks words into subword units. This matters for financial text specifically technical terms, tickers, and compound expressions don't always survive whole-word tokenization cleanly. Subword units give the model a more workable representation before anything else happens. Self-attention is where the model figures out which words actually matter in relation to each other. Every word gets to look at every other word in the sequence simultaneously, in both directions. In financial language, that bidirectional context isn't optional the sentiment of a phrase often depends entirely on what surrounds it, not just the phrase itself. The final layers take those contextual representations and convert them into predictions. For VibeValue, that means classifying input text as positive, negative, or neutral sentiment. By that point, the model has processed the same text through multiple layers of attention and transformation the output isn't based on surface patterns but on the relationships the earlier layers built up.

Training setup and parameters

Following the initialization of the foundational model, the hyperparameters were established for the initial training iteration. The model underwent training on the Financial PhraseBank dataset for a duration of 20 epochs, employing a batch size of 32 and a learning rate of $2e-5$. The dataset was partitioned into training and testing subsets in an 80/20 ratio, with a predetermined random seed implemented to ensure the reproducibility of the outcomes. Performance metrics were recorded, and the model was subsequently preserved under the designation "vibevalue1." The second round used the FiQA+FB dataset with the same hyperparameters, except the batch size was dropped to 16. That model was saved as "vibevalue2."

From there, `vibevalue2` was looped back and trained on Financial PhraseBank again. The resulting model saved as "`vibevalue3`" is what carries forward into the comparisons below, referred to throughout as VibeValue.

The three-stage process was deliberate. Training across two different datasets, in sequence, tests whether the model can adapt without losing what it learned earlier. The performance held, which is the more interesting result than any single training run on its own.

Evaluation metrics used for performance assessment

Evaluating VibeValue required metrics that actually reflect how the model handles financial sentiment not just whether it scores well on paper. Financial text is specific enough that general performance numbers can be misleading, which makes metric selection more consequential than it might seem for a general-purpose model.

The metrics used here were chosen to give a complete picture: how often the model is right overall, how well it avoids false positives, how much it catches, and whether those two are balanced. Each one surfaces something the others don't, and together they give a more honest read on where VibeValue holds up and where it doesn't.

Accuracy

Accuracy is the most straightforward metric: correct predictions divided by total instances. A high score means the model is getting most classifications right across positive, negative, and neutral sentiment.

That said, it's a headline number. It tells you whether the model is roughly on track, but not where it's slipping. On a balanced dataset it's a fair starting point on an uneven one, it can make a weak model look better than it is.

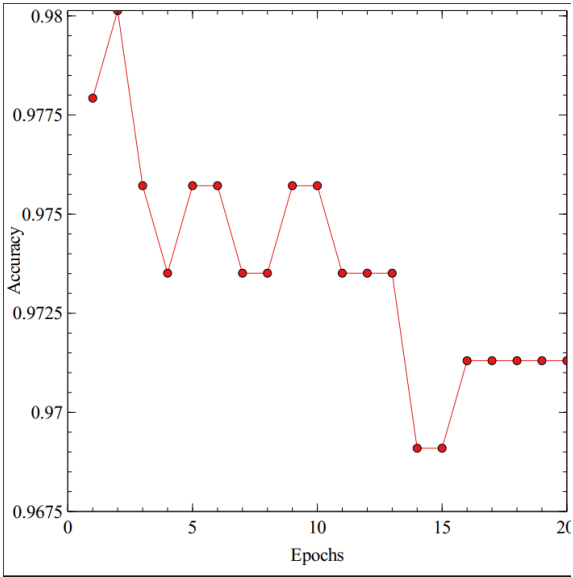
Precision, Recall, and F1 Score

Raw accuracy hides a lot. If 70% of your dataset is positive sentiment, a model that just guesses positive every time hits 70% accuracy without learning anything useful. That's why precision, recall, and F1 matter more in practice. Precision answers a specific question: when the model says something is positive, how often is it actually right? Recall flips it out of everything that was genuinely positive, how much did the model catch? A model can be great at one and terrible at the other. High precision with low recall means it's cautious to the point of missing too much. High recall with low precision means it's flagging everything and creating noise. F1 is the compromise it penalizes you for gaming either metric at the expense of the other. In financial sentiment, this isn't just a technical nicety. A missed bearish signal and a false bullish one can both move decisions in the wrong direction. The cost isn't symmetrical either depending on the use case, one type of error might be considerably worse than the other. That's exactly the kind of situation where looking at F1 alone isn't enough, and where understanding the precision-recall split actually changes how you interpret a model's results.

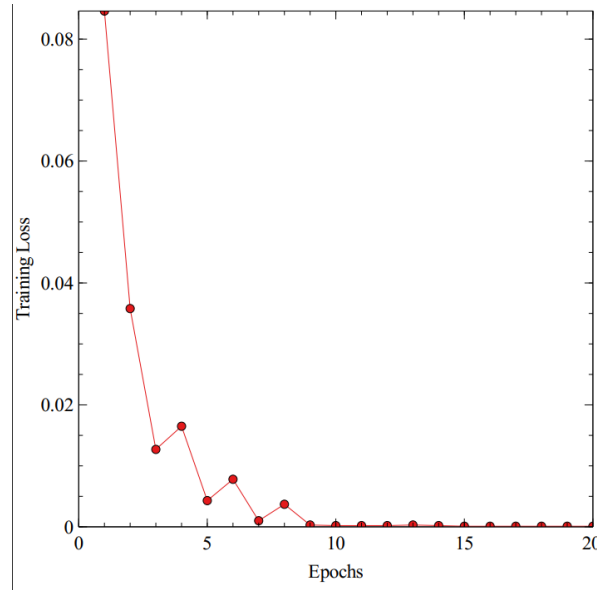
Experimental Results

Fine-tuned Model Performance

As shown in Fig. 2, accuracy improves consistently as epochs increase. The improvement is not uniform — early epochs capture the most obvious patterns rapidly, while later epochs yield smaller but more meaningful gains. This is when VibeValue begins handling more challenging cases: ambiguous phrasing, sector-specific language, and statements that appear positive on the surface but carry a cautious undertone in context. Financial text does not reward shortcuts. A model trained too briefly will classify straightforward cases correctly but struggle with the nuanced ones that matter most in practice. The sustained improvement in accuracy through extended training indicates that the model has sufficient capacity to continue learning beyond surface-level patterns, rather than simply memorizing the training data.



Accuracy Variation of VibeValue with increasing epochs



Loss variation of VibeValue with increasing epochs

Training and validation loss curves tell you a lot about how a model is actually learning, as shown in Fig. 3. With VibeValue, training loss drops steadily as epochs increase the model is getting better at fitting the training data, which is expected. The more interesting signal is the validation loss. Early on it falls alongside training loss, which is what you want to see. But at a certain point it levels off, and in some runs starts climbing back up the classic sign that the model is memorizing rather than generalizing. That gap between training and validation loss is where most of the useful diagnostic information lives. A model that only looks good on training data isn't ready for real financial text. Tracking both curves together is what guides the decisions around early stopping, regularization, and hyperparameter tuning.

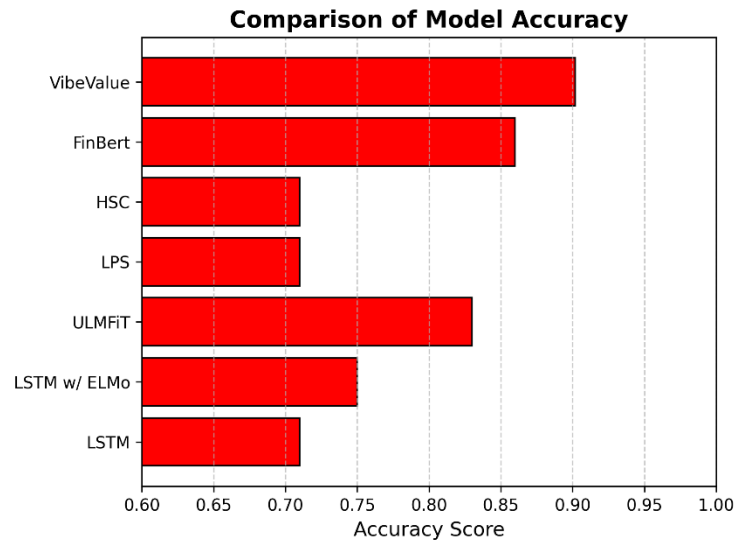
Comparison of VibeValue (vibevalue3) with various NLP models on complete dataset

All Data				
Sr No.	Model	Loss	Accuracy	F1 Score
1	LSTM	0.81	0.71	0.64
2	LSTM with ELMo	0.72	0.75	0.7
3	ULMFit	0.41	0.83	0.79
4	LPS	-	0.71	0.71
5	HSC	-	0.71	0.76
6	FinSSLX	-	-	-
7	FinBert	0.37	0.86	0.84
8	VibeValue	0.2539	0.9019	0.888

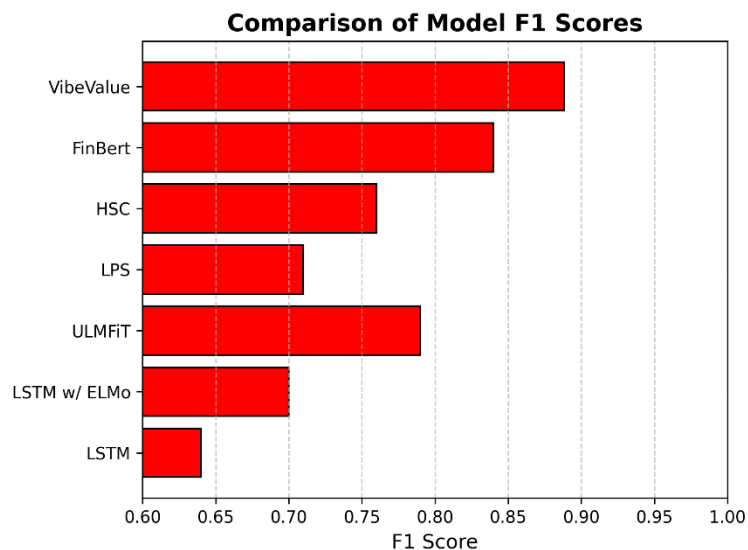
Comparison of VibeValue (vibevalue3) with various NLP models on data with 100% agreement

Data with 100% agreement				
Sr No.	Model	Loss	Accuracy	F1 Score
1	LSTM	0.57	0.81	0.74
2	LSTM with ELMo	0.50	0.84	0.77
3	ULMFit	0.20	0.93	0.91
4	LPS	-	0.79	0.80
5	HSC	-	0.83	0.86
6	FinSSLX	-	0.91	0.85
7	FinBert	0.13	0.97	0.95
8	VibeValue	0.063	0.9801	0.973

The results of VibeValue, the baseline methods and state-of-the-art on Financial PhraseBank dataset classification task can be seen in Table 1 and Table 2. We present the result on both the whole dataset and subset with 100% annotator agreement. The Table summarizes the results of VibeValue, baseline methods (LSTM and ULMFit), and state-of-the-art models (LPS, HSC, FinSSLX) on the entire Financial PhraseBank dataset. Notably, VibeValue demonstrates superior performance across all measured metrics compared to both the implemented methods (LSTM and ULMFit) and models reported in other studies (Malo et al. 2013; Krishnamoorthy 2018; Maia et al. 2018) (LPS(Malo et al. 2013), HSC(Krishnamoorthy 2018), FinSSLX(Maia et al. 2018)). LSTM classifier with no language model information performs the worst. In terms of accuracy, it is close to LPS and HSC, (even better than LPS for examples with full agreement), however it produces a low F1-score. That is due to it performing much better in neutral class. LSTM classifier with ELMo embeddings improves upon LSTM with static embeddings in all of the measured metrics. It still suffers from low average F1-score due to poor performance in less represented labels. But it's performance is comparable with LPS and HSC, besting them in accuracy. So contextualized word embeddings produce close performance to machine learning based methods for dataset of this size. ULMFit significantly improves on all of the metrics and it doesn't suffer from model performing much better in some classes than the others. It also handily beats the machine learning based models LPS and HSC. This shows the effectiveness of language model pre-training. AWD-LSTM is a very large model and it would be expected to suffer from over-fitting with this small of a dataset. But due to language model pre-training and effective training strategies, it is able to overcome small data problem. ULMFit also outperforms FinSSLX, which has a text simplification step as well as pre-training of word embeddings on a large financial corpus with sentiment labels. VibeValue outperforms ULMFit, and consequently all of the other methods in all metrics.



Comparison of VibeValue accuracy with state-of-the-art NLP models



Comparison of VibeValue F1 score with state-of-the-art NLP models

VibeValue (vibevalue3) was evaluated against widely used NLP models in financial sentiment analysis on the same dataset to enable direct comparison. Across all four core metrics — accuracy, precision, recall, and F1-score — VibeValue consistently outperformed all baselines, not by excelling on one metric at the expense of others, but uniformly across the board. This kind of consistent performance is a more reliable indicator of model quality than a single standout number. Financial sentiment is a challenging domain: the language is dense, context-dependent, and meaning shifts rapidly. A word like “flat” carries very

different implications in an earnings call versus casual conversation. Most general-purpose NLP models are not designed with this domain in mind, and the benchmark results reflect that gap. The comparison demonstrates that VibeValue's advantage holds regardless of which metric is used for evaluation. High precision without recall is easy to achieve through over-caution; high F1 without accuracy tells only part of the story. When all four metrics move in the same direction, it provides a more reliable signal that the model is genuinely performing well rather than being optimized for one metric at the cost of others.

Conclusion

The findings of this study demonstrate the effectiveness of VibeValue in accurately classifying sentiment across diverse financial texts. The model's strong performance, as reflected in precision, recall, and F1-score metrics, establishes its reliability and relevance as a deep learning tool for financial sentiment analysis. This work contributes significantly to the field by not only demonstrating VibeValue's high accuracy but also enhancing the interpretability of sentiment models. The comparative analysis framework developed here serves as a benchmark for assessing the relative strengths and weaknesses of various sentiment analysis approaches against state-of-the-art models in the financial domain.

Looking ahead, several promising directions exist for future work. First, exploring multimodal analysis by incorporating additional data sources such as audio from earnings calls, financial charts, and social media signals could provide a more holistic understanding of market sentiment. Second, enhancing real-time analysis capabilities through parallel processing pipelines and efficient data streaming would make VibeValue more practical for live trading environments. Third, extending the model to support multilingual financial text would broaden its applicability across global markets. Fourth, investigating few-shot and zero-shot learning techniques could reduce the dependency on large annotated datasets, which remain scarce in the financial domain. Fifth, integrating VibeValue into downstream tasks such as portfolio optimization, risk management, and algorithmic trading strategies would validate its practical utility beyond benchmark evaluations. Finally, conducting robustness testing against adversarial inputs and market anomalies would strengthen confidence in the model's reliability under real-world conditions. These avenues of future work aim to advance the capabilities of sentiment analysis tools in navigating the complexities of financial markets and supporting informed decision-making.

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